2 CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

The purpose of a literature review is to examine existing knowledge, systems, and research that relate to the proposed study. By exploring global, regional, and local contexts, this chapter highlights how navigation technologies have evolved, their adoption in educational institutions, and the challenges that remain. The review also identifies gaps that justify the development of a customized navigation application for United States International University – Africa (USIU-A).

2.2 Global Perspective

Globally, navigation systems have developed significantly over the past three decades, particularly with the rise of GPS-based services. Applications such as Google Maps, Apple Maps, and Waze are now widely used for real-time route planning and location-based services. These tools have proven effective in urban mobility, tourism, and logistics by offering turn-by-turn navigation, traffic updates, and estimated time of arrival.

Within universities, several institutions in North America and Europe have integrated digital navigation tools into their campus services. For instance, some universities have implemented mobile applications that allow students to search for lecture halls, offices, and libraries, often combined with class schedules or campus events. According to research studies, the implementation of campus-specific navigation systems contributes to improved time management, reduced anxiety among new students, and better utilization of campus resources. Despite this progress, most global solutions remain tailored to outdoor navigation and require significant customization before they can meet the needs of specific institutions.

2.2.1 MIT WhereIs (Massachusetts Institute of Technology, USA)

The Massachusetts Institute of Technology (MIT) developed WhereIs, a digital campus navigation system that has become a benchmark for other universities globally. This system provides an interactive map of the entire MIT campus, including classrooms, laboratories, dormitories, offices, libraries, and service facilities. Users can search for specific buildings or services and generate walking directions to navigate the large and complex campus. The platform integrates detailed building floor plans, which makes it especially useful for new students and visitors unfamiliar with the campus layout. Because MIT attracts a global community of students, staff, and researchers,

the navigation system is widely used by international users and demonstrates how technology can simplify movement within a large academic environment.

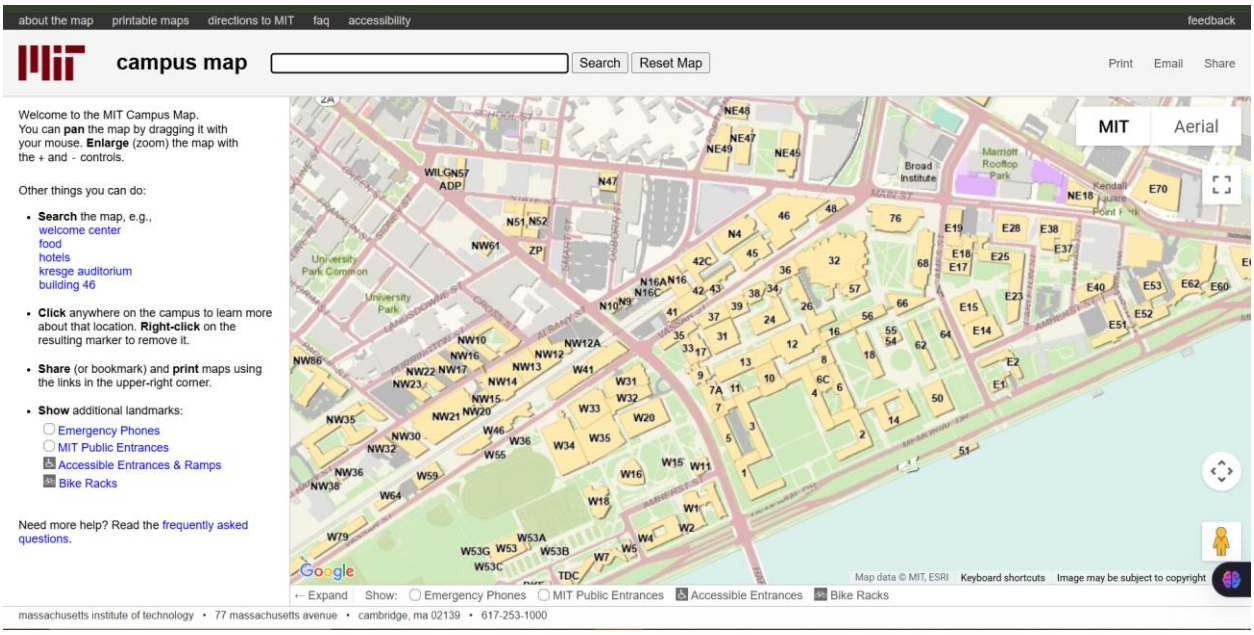

Figure 1: MIT WhereIs system

Figure 1 above represents The MIT WhereIs system whereby it showcases an advanced and well-integrated approach to campus navigation that effectively balances functionality, design, and accessibility. The interface is clear and interactive, allowing users to quickly locate destinations through a simple search or by exploring the digital campus map. Its inclusion of detailed floor plans sets it apart from many other systems, as it provides indoor navigation capabilities that help students and visitors find specific rooms, offices, or facilities within multi-story buildings. This feature is particularly valuable in a complex academic environment like MIT, where buildings are interconnected and large in scale.

In addition to its design features, the platform demonstrates a high level of technical efficiency. Being web-based, it requires no installation, which makes it easy for both first-time visitors and members of the MIT community to access it from any device. The integration of accurate geospatial data enhances the reliability of directions and ensures that the map remains consistent with the physical layout of the campus. Furthermore, its ability to highlight important services—such as libraries, cafeterias, or parking areas—adds functional depth that goes beyond simple route mapping.

The success of the WhereIs platform emphasizes the importance of user-centered design and continuous system updates to maintain accuracy and usability. For the USIU Campus Navigation System, this example underlines key aspects to adopt: clear and interactive interfaces, searchable locations, multi-level mapping, and mobile accessibility. Implementing similar elements can ensure that the system not only guides users effectively but also improves their overall campus experience through convenience and efficiency.

2.2.2 NUS NextBus and Campus Map (National University of Singapore, Singapore)

The National University of Singapore (NUS) has a comprehensive campus navigation system integrated into its NextBus mobile application and interactive campus maps. This system not only provides directions to buildings such as lecture halls, libraries, and administrative offices but also integrates real-time information about campus shuttle buses. Students and visitors are able to check the nearest bus stops, track arrival times, and plan routes that combine walking and campus transport. The NUS system is widely recognized in Asia for its efficiency and for addressing the specific needs of a large campus with multiple transport routes. Its integration of navigation with transport scheduling sets it apart as a global example of how universities can combine location services with real-time updates to improve user experience.

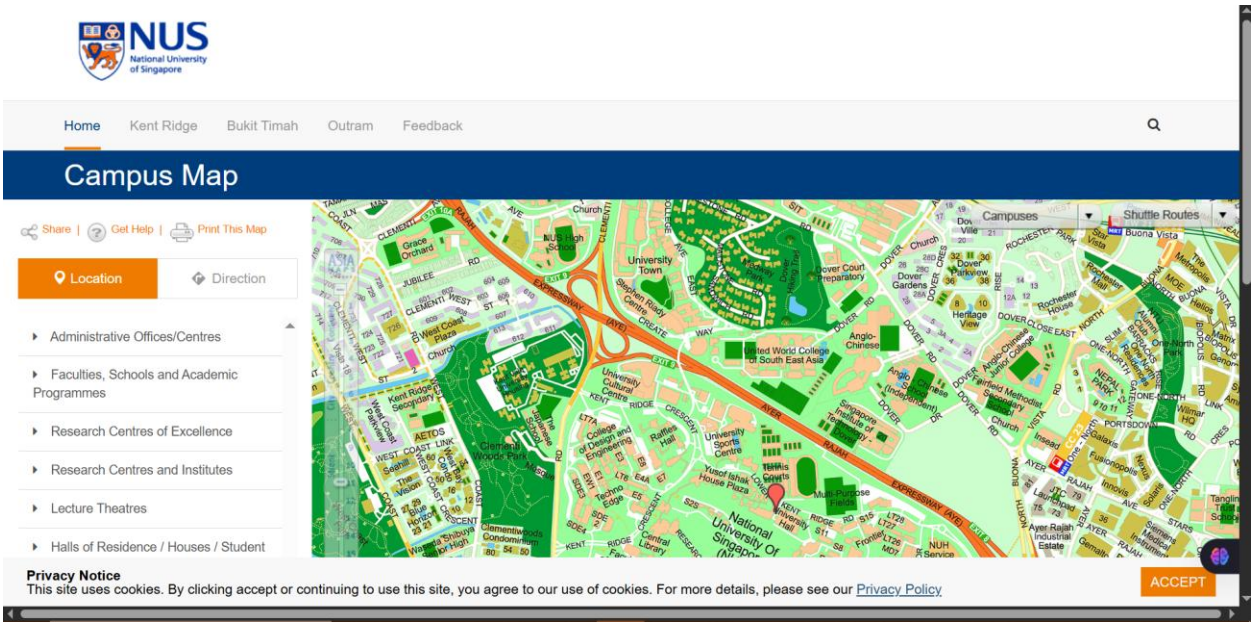

Figure 2:NUS NextBus and Campus Map system

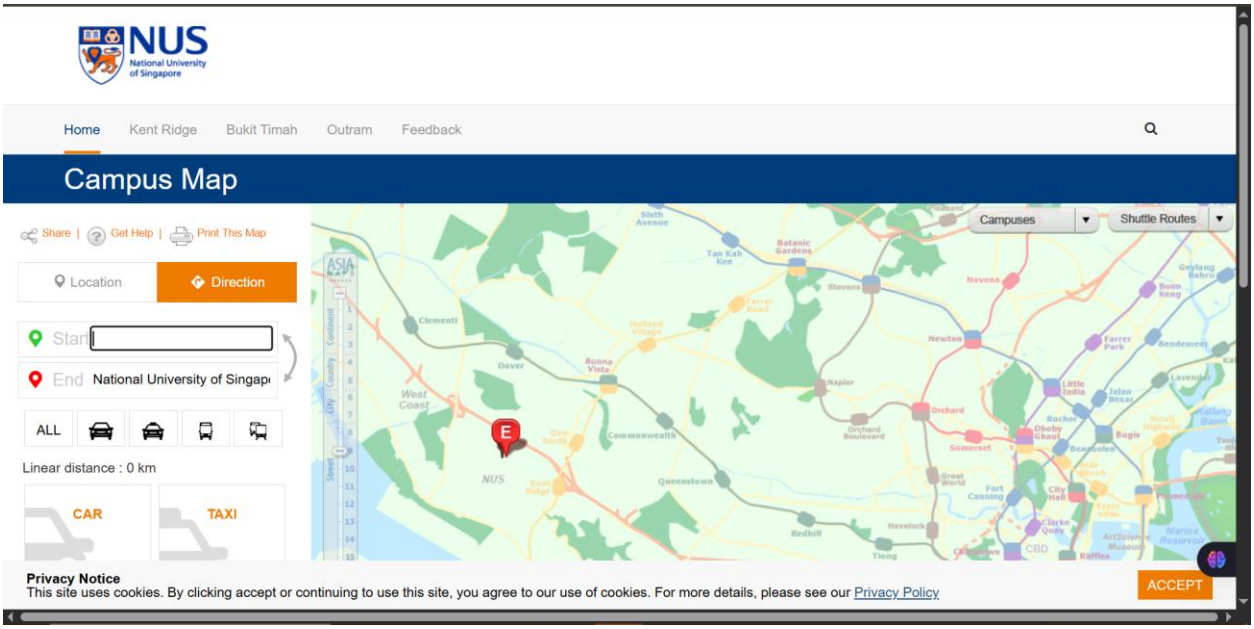

Figure 3: NUS NextBus and Campus Map system

Figure 2 and 3 above illustrate The NUS NextBus and Campus Map system represents a sophisticated blend of navigation and transportation management designed to address the challenges of movement within a large and geographically dispersed campus. The interface effectively integrates building directions with real-time shuttle tracking, providing users with both pedestrian and vehicular route options. This dual-function approach allows students, staff, and visitors to plan their movements more efficiently, especially during peak hours or when transitioning between distant faculties. The incorporation of live bus arrival updates demonstrates how smart technology can be used to minimize waiting times and reduce congestion around key bus stops.

In terms of usability, the platform emphasizes clarity and responsiveness. The map interface is visually organized, displaying color-coded bus routes, building icons, and walking paths to help users navigate intuitively. Its integration with mobile devices ensures that users can access updated information anywhere on campus, making it a convenient and dynamic tool. The system also supports accessibility through multilingual options and location-based services that enhance inclusivity for international students and visitors.

The innovation seen in the NUS system highlights the potential of combining real-time data and spatial mapping for improved campus mobility. By merging navigation with transport information,

NUS demonstrates a holistic approach that goes beyond static maps. For the USIU Campus Navigation System, this serves as a valuable model to emulate—particularly the inclusion of dynamic features such as live location tracking, estimated time of arrival (ETA) functions, and route optimization. Incorporating such elements would make the USIU system not only a navigation aid but also an intelligent mobility platform that enhances convenience, time management, and user satisfaction across the campus community.

2.3 Regional Perspective

In the African context, digital transformation is steadily expanding due to increased smartphone penetration and internet accessibility. However, the integration of smart navigation systems in universities remains limited. Most institutions continue to rely on printed maps, signage, or orientation programs to guide new students and visitors. While some African universities have adopted e-learning platforms and digital communication tools, fewer have invested in specialized navigation systems.

Research shows that one of the main barriers to the adoption of campus-specific navigation applications in Africa is the cost of implementation and maintenance, as well as limited technical expertise in some institutions. Nonetheless, as the concept of "smart campuses" gains ground, more universities are beginning to explore the integration of digital solutions for navigation, safety, and administrative services. This provides an opportunity for USIU-A, a technology-driven institution, to pioneer a campus navigation application that can serve as a model for other universities in the region.

2.3.1 University of Cape Town (UCT) Campus Maps, South Africa

The University of Cape Town (UCT), one of Africa's leading universities, has developed detailed campus maps to guide students, staff, and visitors across its multiple campuses. The UCT campus is spread across several locations, including Upper, Middle, and Lower Campuses, which makes navigation challenging for newcomers. To address this, UCT provides online maps and downloadable PDFs that highlight academic buildings, student residences, libraries, and service centers. In addition, signage and map boards placed around the campus complement the digital tools. While not as technologically advanced as mobile navigation apps like MIT's WhereIs,

UCT's system is a strong regional example that demonstrates how African universities are adopting digital mapping solutions to enhance accessibility and orientation.

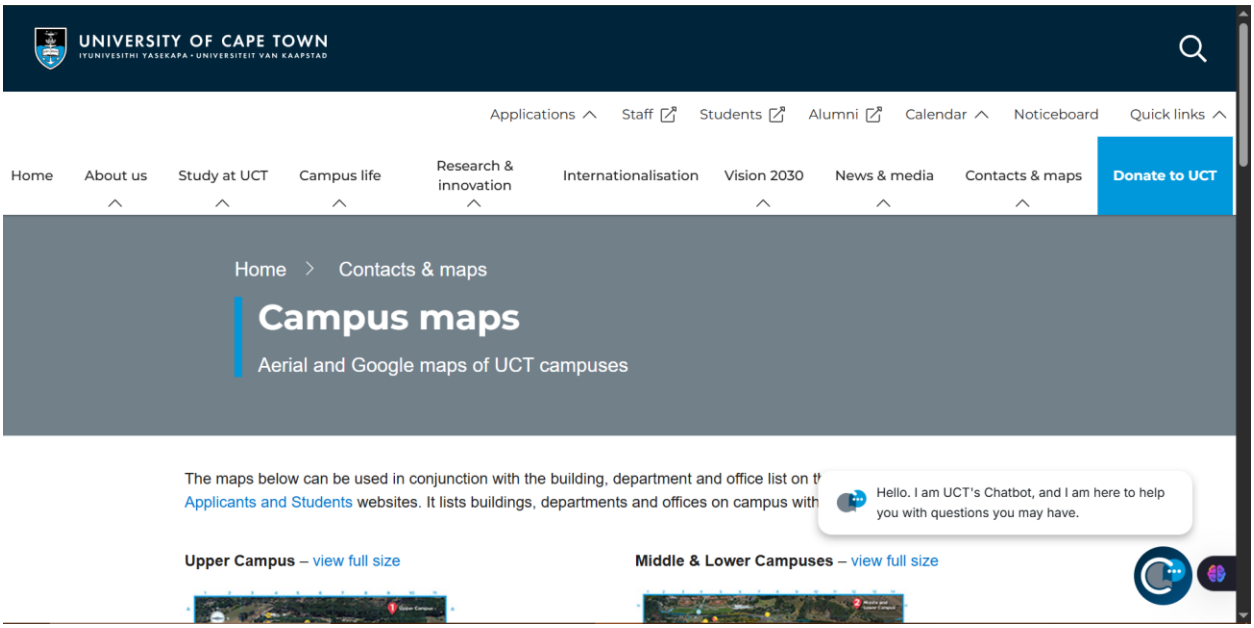

Figure 4: UCT Campus Maps system

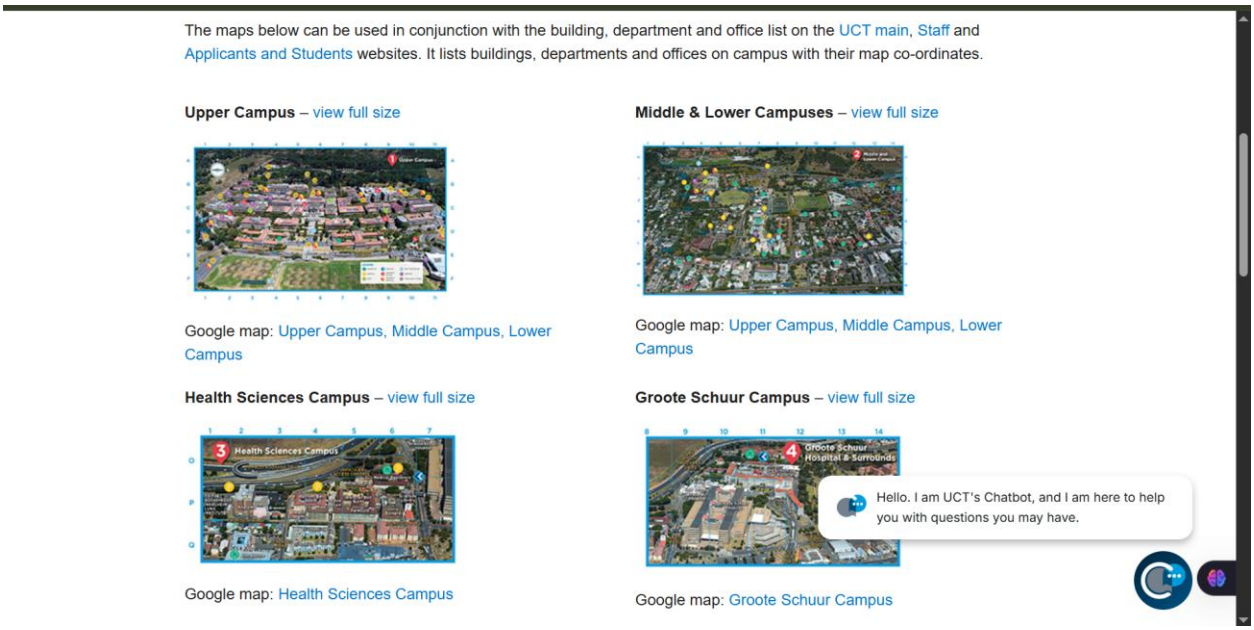

Figure 5: UCT Campus Maps system

Figure 4 and 5 above show The UCT Campus Maps system offers a practical and regionally relevant approach to campus navigation by emphasizing accessibility and clarity over complexity.

The combination of digital and physical navigation tools ensures that users—especially first-year students and visitors—can easily locate facilities across the university's expansive and geographically tiered campuses. The online and downloadable map formats allow for offline access, which is particularly beneficial in contexts where mobile internet connectivity may be inconsistent. This approach reflects a user-centered design philosophy that prioritizes inclusivity and adaptability within an African setting.

Although UCT's system may not incorporate advanced technologies such as real-time GPS tracking or interactive indoor maps, it succeeds in presenting well-structured and clearly labeled information that meets the needs of its users. The integration of physical map boards across campus further supports the digital platform, ensuring navigation support for individuals who may not have smartphones or prefer visual references. The color-coded layout and categorized icons used in UCT's maps make them visually straightforward and easy to interpret, thereby minimizing confusion in a multi-campus environment.

From a regional perspective, UCT's navigation system demonstrates how universities can effectively blend low-cost digital solutions with on-site infrastructure to enhance campus wayfinding. This hybrid model offers valuable insights for institutions operating in similar contexts where resource constraints may limit the deployment of high-end digital platforms. For the USIU Campus Navigation System, the UCT approach highlights the importance of accessibility, clarity, and multi-format usability. Integrating both online and offline navigation features such as downloadable maps or strategically placed map boards could make the USIU system more inclusive and reliable, even in cases of network disruptions or limited device access.

2.3.2 University of Lagos (UNILAG) Campus Navigator, Nigeria

The University of Lagos (UNILAG) has implemented a semi-digital campus navigation solution that combines web-based maps and a mobile-accessible guide to help students and visitors locate academic departments, lecture halls, and facilities. The system uses geotagged locations and route indicators to support easy movement within the university's expansive environment. While the platform lacks real-time GPS integration, it represents a growing effort among African universities to digitize campus navigation and improve user experience. UNILAG's initiative illustrates the regional shift towards adopting campus mapping technologies in higher education institutions across Africa.

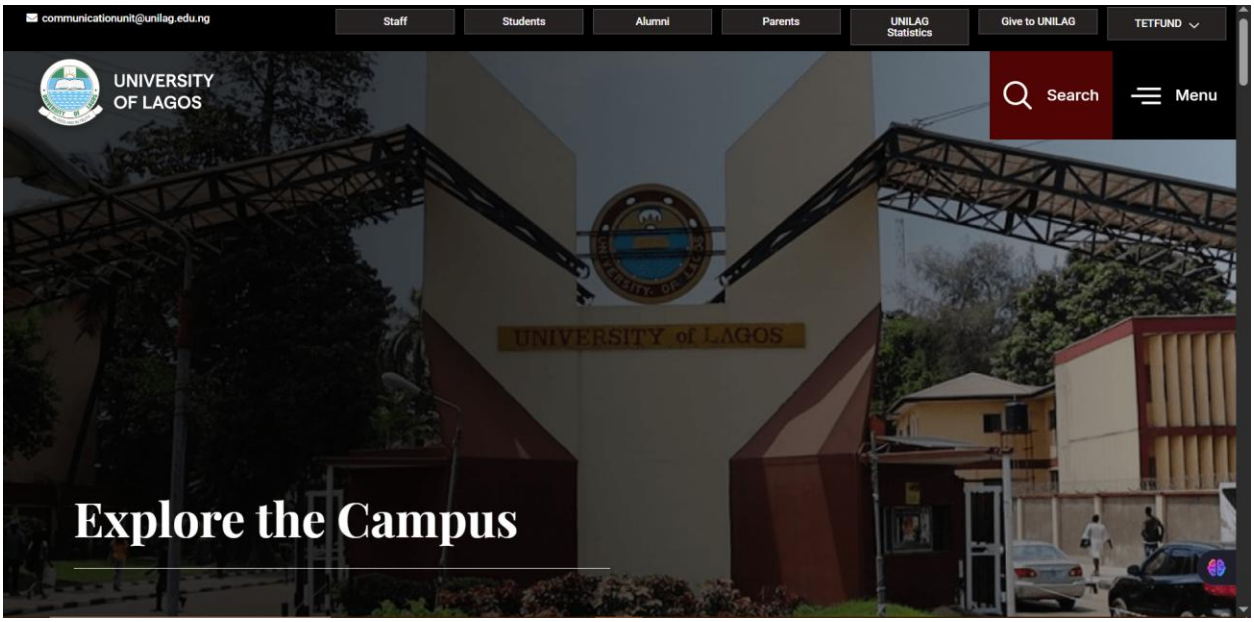

Figure 6:UNILAG Campus Navigator home page

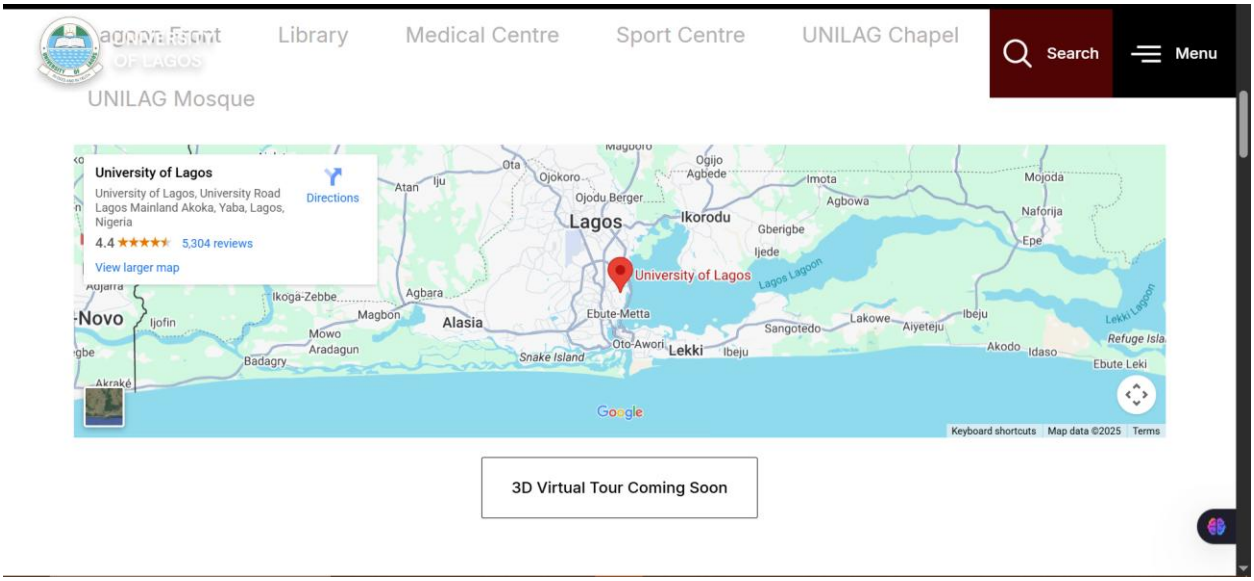

Figure 7: UNILAG Campus Navigator

Figure 6 and 7 above show The UNILAG Campus Navigator exemplifies the progressive digital transformation taking place within African higher education institutions. Its integration of web-based maps with geotagged points offers users an organized overview of the university's large and complex layout. The use of route indicators allows students and visitors to visualize directions between key locations such as lecture halls, administrative offices, and residential areas. Although it operates without real-time GPS tracking, the system still provides a strong foundation for

location-based services within the university context. This semi-digital approach demonstrates how institutions with limited technological infrastructure can still leverage accessible web tools to improve campus mobility.

The design of the UNILAG navigation system focuses on simplicity and functionality. Its clear labeling, clickable icons, and structured layout enable users to identify destinations without confusion. The platform's mobile compatibility ensures that users can access maps on smartphones, which aligns with the growing reliance on mobile devices among university students. This responsiveness makes it convenient for on-the-go users navigating the large UNILAG campus. Additionally, the inclusion of essential facilities such as libraries, cafeterias, hostels, and parking spaces enhances the system's practicality for both new and continuing users.

From a regional perspective, UNILAG's effort marks an important step toward digital modernization in African universities. It reflects how institutions are gradually embracing mapping technologies to improve user experience and campus accessibility. For the USIU Campus Navigation System, the UNILAG model offers insights into balancing innovation with feasibility. While advanced features such as live GPS tracking and real-time updates may come later, starting with geotagged locations, interactive icons, and mobile adaptability would create a strong and sustainable foundation. By building upon such regional innovations, USIU can design a navigation platform that is both technologically progressive and contextually relevant to African university environments.

2.4 Local Perspective

In Kenya, several universities are gradually embracing digital transformation to enhance student experiences and operational efficiency. While many institutions still rely on physical maps and signage, some have begun adopting digital campus navigation tools to improve orientation for new students and visitors. These local efforts serve as a foundation for USIU-A's proposed campus

navigation system, which aims to integrate modern geolocation features, building information, and interactive guidance.

2.4.1 University of Nairobi (UoN) Wayfinding Maps, Kenya

The University of Nairobi (UoN) has developed online and printable wayfinding maps that display the layout of its main and satellite campuses. These maps provide detailed information on the

locations of lecture halls, faculty buildings, administrative offices, and student service areas. While the system is primarily static and lacks real-time GPS tracking, it plays a crucial role in helping users navigate the university's vast environment. UoN's approach highlights the local need for more interactive and digital navigation systems that can provide real-time guidance for both students and visitors.

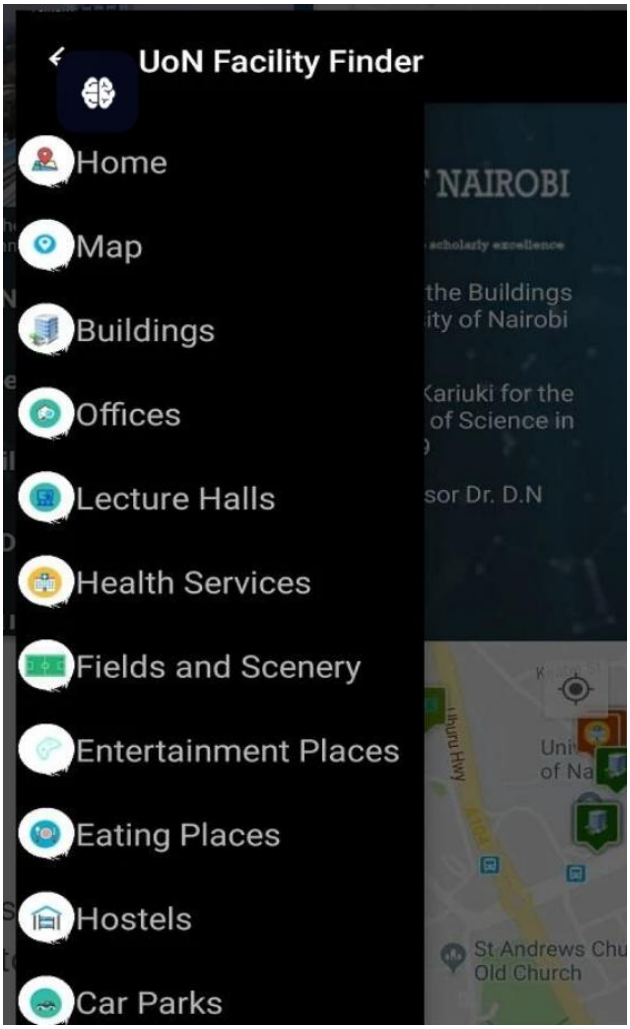

Figure 8: The University of Nairobi's Wayfinding Maps system

Figure 9: The University of Nairobi's Wayfinding Maps system

Figure 8 and 9 above show The University of Nairobi's Wayfinding Maps system demonstrates an important local effort toward structured campus navigation and spatial organization. By offering both online and printable map formats, the university ensures accessibility for a wide range of users, including those with limited internet access or older mobile devices. The maps are well-labeled and organized according to faculties, lecture halls, and key administrative areas, making them useful for first-time visitors, new students, and staff. The inclusion of both main and satellite campuses also reflects the university's awareness of its broad geographical footprint and the need for unified navigation support across locations.

Despite being static and lacking dynamic GPS or interactive routing features, the system effectively fulfills its core objective: to provide clear visual references that help users locate essential facilities within the institution. Its simple, graphic-oriented design promotes quick interpretation, which is particularly beneficial in a setting with high pedestrian traffic and multiple entry points. Moreover, the use of standardized icons and color codes across the maps enhances consistency and reduces the likelihood of confusion for users unfamiliar with the campus layout.

From a local perspective, UoN's initiative highlights both the progress and limitations of digital mapping within Kenyan universities. It underscores the ongoing transition from static representations to more interactive, data-driven solutions. For the USIU Campus Navigation System, the UoN model offers valuable lessons in prioritizing clarity, accessibility, and comprehensiveness before implementing advanced features. Building upon this foundation, USIU can incorporate interactive elements such as real-time positioning, voice-assisted directions, and building search filters to create a more advanced and engaging platform. Ultimately, this evolution from static maps to smart navigation technology represents the next step in improving campus mobility and user experience within Kenyan higher education institutions.

2.4.2 Strathmore University Smart Campus Map, Kenya

Strathmore University has adopted a smart campus approach that integrates digital mapping into its student information system. The system allows users to locate lecture rooms, departments, and essential facilities through an interactive interface accessible on mobile and desktop platforms. The university's initiative reflects a growing local trend toward integrating technology in campus management and student services. However, the system is limited in terms of real-time navigation and GPS-based guidance, creating an opportunity for USIU-A's campus navigation system to offer a more advanced, location-aware solution tailored to user movement within the university.

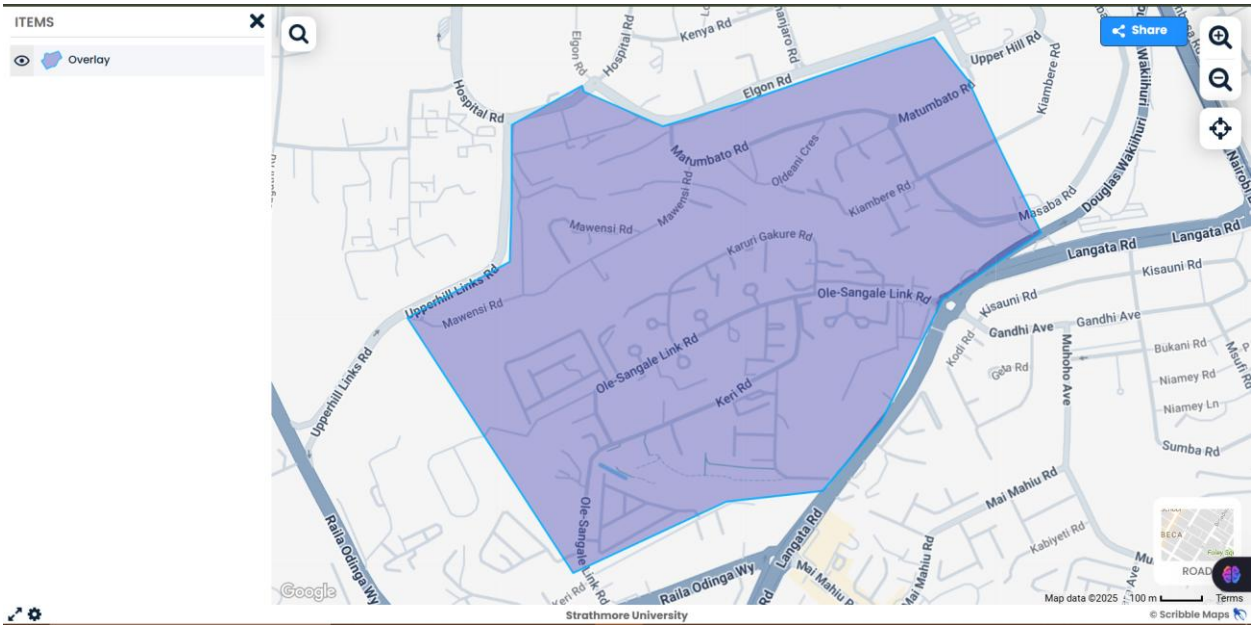

Figure 10: The Strathmore University Smart Campus Map

Figure 10 above shows The Strathmore University Smart Campus Map represents one of the most forward-looking digital navigation initiatives within Kenya's higher education landscape. By integrating its mapping platform into the broader student information system, Strathmore demonstrates how technology can be embedded into academic and administrative functions to enhance convenience and connectivity. The map's interactive interface allows users to search for and identify buildings, departments, and service areas with ease. Its compatibility with both desktop and mobile devices further strengthens its accessibility, ensuring that users can obtain navigation support at any time and from any location. This integration of mapping and institutional data positions Strathmore as a pioneer in developing a more connected and technology-driven learning environment.

However, while the system's structure is user-friendly and technologically advanced compared to many local institutions, its static nature limits the full potential of location-based interaction. The absence of real-time navigation and GPS functionality means that users must rely primarily on visual orientation rather than live guidance. This restricts the system's usefulness in providing turn-by-turn directions or assisting new students and visitors in locating precise destinations across the campus. Despite these constraints, the platform remains an important milestone in Kenya's digital campus transformation journey, signaling a shift toward smart campus management systems that prioritize efficiency, accessibility, and digital inclusion.

For the USIU Campus Navigation System, the Strathmore model provides an excellent foundation to build upon. It illustrates the benefits of system integration, cross-platform accessibility, and intuitive design. However, it also reveals opportunities for innovation—specifically through real-time GPS guidance, route optimization, and interactive feedback features. By incorporating these advancements, the USIU system could go beyond static mapping to create a dynamic, location-aware solution that not only enhances campus mobility but also contributes to a smarter and more connected university ecosystem.

2.5 Strengths and Weaknesses of the Selected Systems

Section 2.5 presents a critical evaluation of the strengths and weaknesses of selected campus navigation systems. By analyzing both local and international solutions, this section aims to highlight the core functionalities, benefits, and limitations of each system to provide insights that inform the design and development of a more efficient, context-appropriate navigation platform for university environments.

Table 1: Strengths and Weaknesses of the Selected Campus Navigation Systems

Table 1: Strengths and weaknesses of selected systems

System	Strengths	Weaknesses
MIT WhereIs (USA)	- Comprehensive digital mapping of all campus facilities - Real-time directions and interactive search functionality - Integration of detailed building floor plans - Accessible via web and mobile devices - High reliability and user-friendly interface	- Requires strong internet connectivity - High maintenance cost for large campuses - Not customizable for smaller institutions

NUS NextBus & Campus Map (Singapore)	- Combines navigation with real-time shuttle tracking - Efficient route planning and transport integration - Multi-device compatibility (mobile and web) - Supports live bus arrival notifications	- Focused on campuses with internal transport systems - Complex system architecture may not be ideal for smaller universities
University of Cape Town (UCT) Campus Maps (South Africa)	- Detailed online and downloadable campus maps - Clear labeling of facilities and buildings - Supports multi-campus navigation - Complements digital maps with physical signage	- Lacks real-time GPS tracking - No mobile application for interactive navigation - Manual map updates required
University of Lagos (UNILAG) Campus Navigator (Nigeria)	- Combines static and semi-interactive campus maps - Geotagged locations for key buildings and facilities - Easy to access via mobile and web portals - Supports student orientation and visitor guidance	- Limited real-time positioning - No integration with navigation APIs - Dependent on regular manual updates
University of Nairobi (UoN) Wayfinding Maps (Kenya)	- Provides web-based and physical maps for navigation - Covers multiple campuses and key buildings - Simple and accessible to all users	- Static design with no real-time updates - Lacks mobile app integration - Requires manual map management

	- Useful during student admissions and events	
Strathmore University Smart Campus Map (Kenya)	- Interactive digital map integrated into university systems - Supports location-based searches for rooms and offices - Modern, mobile-responsive interface - Reflects "smart campus" digital innovation efforts	- Lacks real-time GPS tracking and routing - No turn-by-turn navigation - Limited scalability for larger campuses
Proposed Campus Navigation System (USIU-Africa)	- Built specifically for academic environments - Real-time navigation using GPS and map integration - Centralized access to building information and landmarks - Mobile-friendly with simple, intuitive design - Supports both students and visitors - Minimal training required	- Requires stable campus internet connectivity - Initial setup and map digitization may take time - Limited coverage outside university grounds

Analysis

Table 1 above presents the strengths and weaknesses of various existing campus navigation systems, ranging from global solutions such as *MIT WhereIs* and *NUS NextBus*, to regional platforms like UCT Campus Maps and UNILAG Campus Navigator, and local initiatives including UoN Wayfinding Maps and Strathmore Smart Campus Map. Based on this evaluation, clear gaps and opportunities are visible in the current campus navigation landscape.

Global systems are highly advanced, featuring real-time mapping, integration with transport networks, and sophisticated data visualization. However, their complexity, high cost, and need for continuous technical support make them less practical for smaller or mid-sized institutions. Conversely, regional and local systems are simpler and more accessible, offering cost-effective solutions and localized relevance but lacking in real-time functionality, automation, and scalability.

This discrepancy highlights the need for a balanced, context-aware campus navigation solution that integrates the best features of global systems (like real-time mapping and data accuracy) while maintaining the simplicity and affordability of local platforms.

The proposed USIU-Africa Campus Navigation System aims to fill this gap by delivering a GPS-enabled, mobile-responsive platform designed for university environments. It emphasizes real-time location tracking, user-friendly interfaces, and accessibility for students, visitors, and staff. Unlike static maps or externally hosted solutions, this system will be modular, easy to update, and tailored to the specific layout and infrastructure of the campus.

By focusing on interactivity, accessibility, and localized design, the system will transform the campus experience—simplifying navigation, enhancing accessibility, and reducing time spent searching for facilities. This positions USIU-Africa as a leader in smart campus innovation within the region.

2.6 Conclusion

The literature reviewed demonstrates that while navigation technology has advanced globally, localized applications for universities are still underdeveloped, particularly in Africa. By addressing this gap, the proposed system will not only improve campus accessibility but also position USIU-A as a leader in adopting innovative, technology-driven solutions.